

CCNA 200-301 Lab Mastery Roadmap

This document outlines the 35 essential labs required to master the CCNA curriculum. Use this as a checklist to track your hands-on progress.

#	What we're working on	Level	Setup	What's the goal?	Essential Commands
01	Basic IP Addressing	Easy	2 PCs → 1 Switch	Just get them talking with IPs!	<code>ip address</code> , <code>no shut</code> , <code>ping</code>
02	Switching Basics	Easy	4 PCs → 1 Switch	Let's see how the switch learns!	<code>show mac address-table</code>
03	Router Interfaces	Easy	1R → 1SW → 2 PCs	Getting networks to cross the router!	<code>show ip int brief</code> , <code>description</code>
04	Double the Switches	Easy	2SW → 4 PCs	Connecting gear and checking neighbors!	<code>show cdp neighbors</code>
05	VLANs & SVIs	Easy	2SW, VLAN 10/20	Grouping your devices together!	<code>vlan [id]</code> , <code>switchport mode access</code>
06	STP Basics	Easy	2SW Redundant	Making sure we don't crash the network!	<code>show spanning-tree</code>
07	Inter-VLAN (ROAS)	Medium	1R → 1SW Trunk	Routing between those VLANs!	<code>int [id].[sub]</code> , <code>encap dot1Q</code>
08	Static Routing	Medium	3 Routers	Telling the routers where to go!	<code>ip route [net] [mask] [next-hop]</code>
09	RIP v2	Medium	4 Routers	Letting the routers talk to each other!	<code>router rip</code> , <code>version 2</code>
10	OSPF Single Area	Medium	5 Routers	Building a solid backbone!	<code>router ospf</code> , <code>network [wildcard]</code>
11	VTP	Medium	3 Switches	Spreading VLAN info automatically!	<code>vtp mode</code> , <code>vtp domain</code>
12	DHCP Server	Medium	R or L3 Switch	Handing out IPs like a pro!	<code>ip dhcp pool</code> , <code>default-router</code>
13	Standard ACLs	Medium	Router + LANs	Keeping the unwanted traffic out!	<code>access-list [id] permit/deny</code>
14	Extended ACLs	Medium	Router + LANs	Getting specific with what we block!	<code>access-list 100 permit tcp eq 80</code>
15	Static NAT	Medium	LAN → WAN	Mapping your private stuff to the web!	<code>ip nat inside source static</code>
16	PAT (Overload)	Medium	LAN → WAN	Sharing one IP with everyone!	<code>ip nat inside source list overload</code>
17	Serial WAN Links	Pro	2R Point-to-Point	Connecting those old-school links!	<code>encapsulation ppp</code> , <code>clock rate</code>
18	Multi-Router WAN	Pro	3 Routers	Handling paths across the globe!	<code>show ip route static</code>
19	RIP Over WAN	Pro	4 Routers	Routing over those long-distance lines!	<code>passive-interface default</code>
20	OSPF Over WAN	Pro	5 Routers	Fine-tuning OSPF for the WAN!	<code>ip ospf network point-to-point</code>
21	Port Security	Pro	3 Switches	Locking down your physical ports!	<code>switchport port-security</code>
22	Security & ROAS	Pro	R-on-a-stick	Keeping your VLANs safe!	<code>show access-lists</code>
23	DHCP Security	Pro	R + LANs	Making sure DHCP stays reliable!	<code>ip dhcp excluded-address</code>

24	NAT Fixes	Pro	R + LAN + WAN	Figuring out why NAT isn't working!	debug ip nat
25	Multi-area OSPF	Pro	6 Routers	Scaling OSPF to the big leagues!	area [id] range [prefix]
26	HSRP Gateway	Pro	2R Redundant	Making sure the internet never goes down!	standby [group] ip , priority
27	Default Routes	Pro	R + WAN	Sharing the way out to everyone!	default-information originate
28	Full Integration	Pro	Mixed Hardware	Bringing it all together in one place!	show vlan brief , show ip route
29	The Service Combo	Pro	R + LAN + WAN	A little bit of everything at once!	show ip dhcp binding
30	STP + HSRP	Pro	Full Redundancy	Double protection for your network!	show standby brief
31	Redistribution	Pro	5 Routers	Mixing and matching routing protocols!	redistribute rip subnets
32	Hardened NAT	Pro	R + WAN	NAT with a serious security layer!	show ip nat translations
33	Practice Lab A	Pro	Full Mesh	Testing your skills on a huge network!	copy run start
34	Practice Lab B	Pro	Mixed Topology	Seeing if you can find the problems!	traceroute , telnet
35	The Final Exam	Pro	Full Curriculum	Showing off everything you've learned!	ssh -l [user] [ip]

Generated for the CCNA 200-301 Mastery Path.